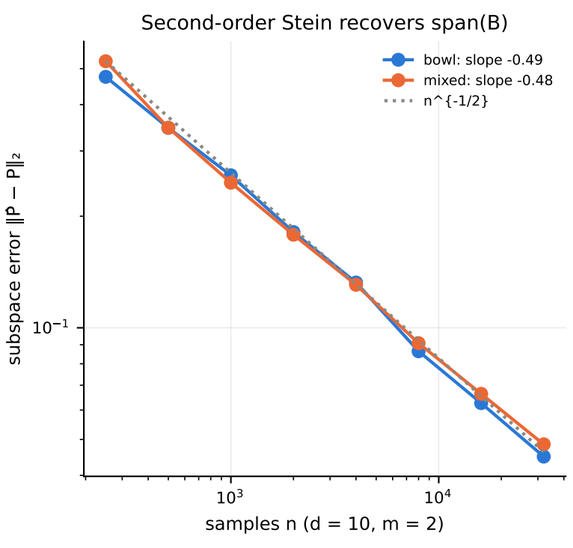
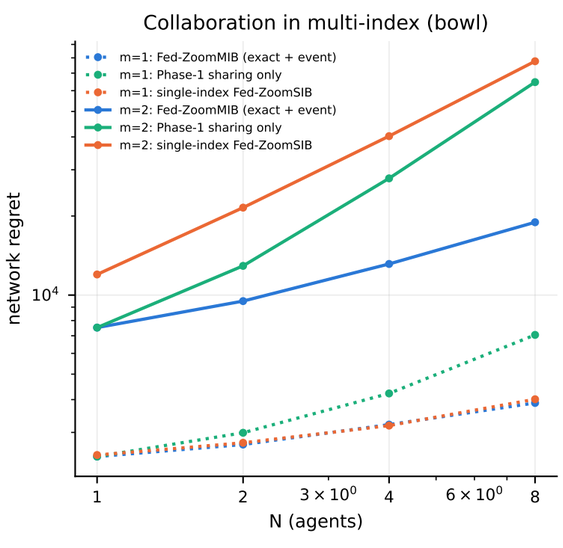
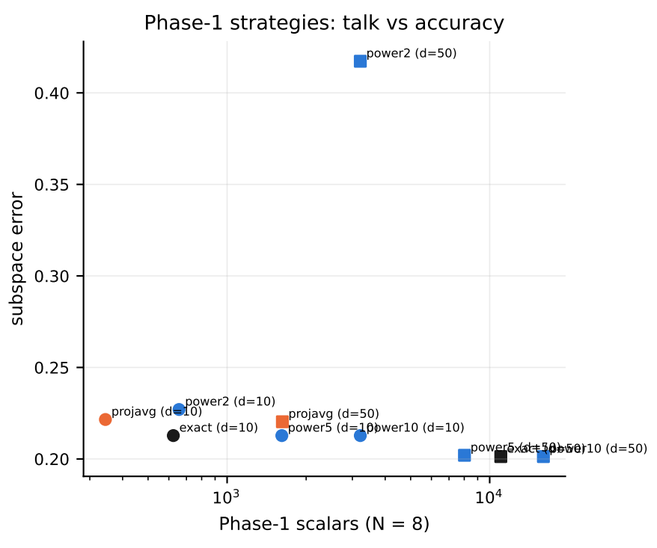
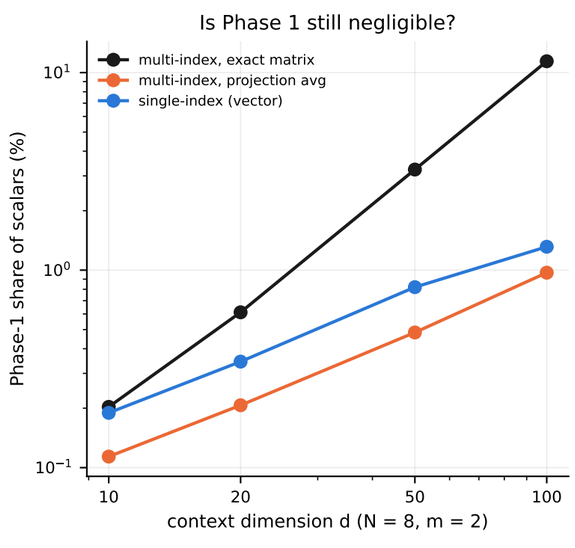

Second-order Stein recovers span(B)
bowl: slope -0.49
mixed: slope -0.48
n^{-1/2}
subspace error $\|P - \hat{P}\|_2$
10^{-1}
10^3
10^4
samples n (d = 10, m = 2)


Collaboration in multi-index (bowl)
m=1: Fed-ZoomMIB (exact + event)
m=1: Phase-1 sharing only
m=1: single-index Fed-ZoomSIB
m=2: Fed-ZoomMIB (exact + event)
m=2: Phase-1 sharing only
m=2: single-index Fed-ZoomSIB
network regret
10^4
1
2
3×10^0
4
6×10^0
8
N (agents)


Phase-1 strategies: talk vs accuracy
power2 (d=50)
subspace error
0.40
0.35
0.30
0.25
0.20
projavg (d=10)
power2 (d=10)
exact (d=10)
projavg (d=50)
power5 (d=10)
power10 (d=10)
power5 (d=50)
power10 (d=50)
10^3
10^4
Phase-1 scalars (N = 8)


Is Phase 1 still negligible?
multi-index, exact matrix
multi-index, projection avg
single-index (vector)
Phase-1 share of scalars (%)
10^1
10^0
10^{-1}
10
20
50
100
context dimension d (N = 8, m = 2)